Computational Thinking

Thinking skill	
Decomposition	1. *Create the Plant class & the Zombie class* 2. *Give each objects attributes (like name, health, damage and distance* 3. *Create attack and movement methods* 4. *Create a loop that controls each turn* 5. *Check whether a plant or a zombie has been defeated* 6. *Determines the winner*
Pattern recognition	1. *Every plant takes 1 hit per turn* 2. *Plant and Zombie loses health when attacked* 3. *A defeated Plant or Zombie can no longer attack* 4. *The turn-based sequence will continue until someone wins*
Abstraction	Plant: 1. Name 2. Health 3. Damage Zombie: 1. Name 2. Health 3. Damage 4. Distance
Algorithm design	1. *Create two plants and one zombie.* 2. *Set their health, damage, and zombie distance.* 3. *Start the game loop.* 4. *Each living plant attacks the zombie.* 5. *If the zombie's health reaches 0, plants win.* 6. *If the zombie is still alive and distance > 0, move the zombie closer.* 7. *If the zombie reaches distance 0, it attacks the first living plant.* 8. *If the first plant is defeated, the zombie attacks the second plant on later turns.* 9. *Repeat until either:* *Zombie health ≤ 0 → Plants win* *Both plants' health ≤ 0 → Zombie wins.*

Class UML Diagram

